The code to generate a basic heat map with Makie is given below:

```
using Makie
using AbstractPlotting: hbox, vbox

 data = rand(50, 100)
 p1 = heatmap(data, interpolate = true)
 p2 = heatmap(data, interpolate = false)
 t = Theme(align = (:left, :bottom), raw = true, camera =
campixel!)
 title1 = text(t, "Interpolate = true")
 title2 = text(t, "Interpolate = false")
 s = vbox(
     hbox(p1, title1),
     hbox(p2, title2),
 )
```

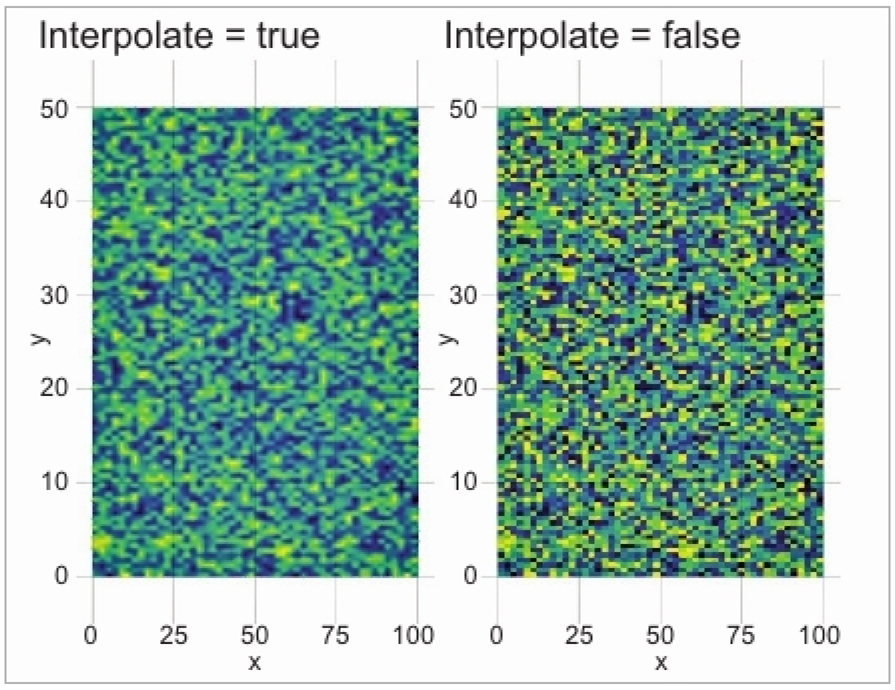

Figure 6: Heat map with Makie

The heat map generated with the above code is shown in Figure 6.

The Makie code blocks shown above are from the official documentation page: *http://makie.juliaplots.org/stable/basic-tutorials.html#Tutorial-1.*

Plots

The Plots package has been built by Thomas Breloff. Plots is an advanced visualisation library with many features. The major design goals of Plots, as specified in the official documentation, are to be:

- Powerful
- Lightweight
- Intuitive
- Consistent
- Concise

The Plots package is powerful and at the same time, it is simple to understand and build the plots with it. An example is shown below [source: *http://docs.juliaplots.org/latest/*]:

```
# load a dataset

using RDatasets
iris = dataset("datasets", "iris");

# load the StatPlots recipes (for DataFrames) available via:
# Pkg.add("StatPlots")
using StatPlots

# Scatter plot with some custom settings
@df iris scatter(:SepalLength, :SepalWidth, group=:Species,
        title = "My awesome plot",
        xlabel = "Length", ylabel = "Width",
        m=(0.5, [:cross :hex :star7], 12),
        bg=RGB(.2,.2,.2))

# save a png
png("iris")
```

Figure 7: Sample plot built using Plots

We can build plots of plot objects with the help of Layouts. A simple code to generate the 2x2 matrix of plots is shown in the following code block. The corresponding output is given in Figure 8.

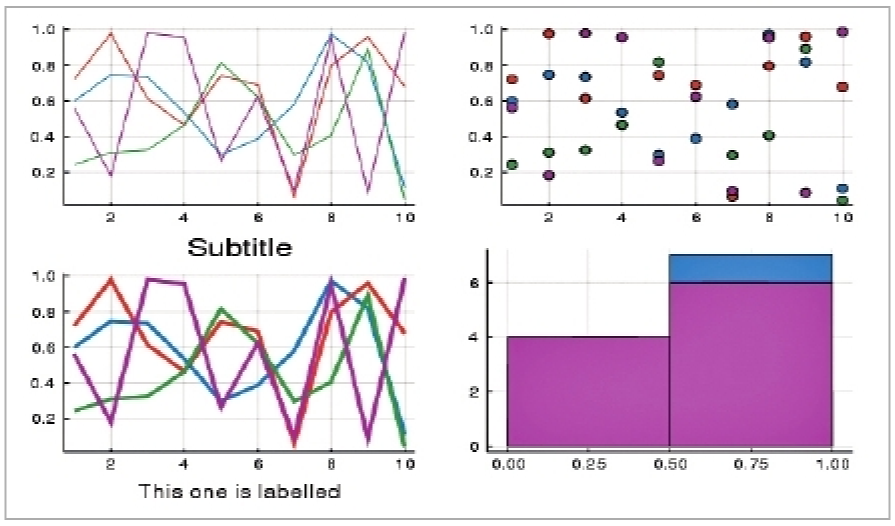

Figure 8: Sub-plots with Plots

```
p1 = plot(x,y) # Make a line plot
p2 = scatter(x,y) # Make a scatter plot
p3 = plot(x,y,xlabel="This one is labelled",lw=3,title="Sub
title")
```